

NATIONAL JUNIOR COLLEGE, SINGAPORE
Senior High 2
Preliminary Examination
Higher 2

CANDIDATE
NAME

BIOLOGY
CLASS

2BI2__ / 2IPBI2__

REGISTRATION
NUMBER

BIOLOGY

9648/02

Paper 2

1 September 2014

2 hours

Additional Materials: Answer Paper and Cover Page

READ THESE INSTRUCTIONS FIRST

Write your Biology class, registration number and name on all the work you submit.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use paper clips, highlighters, glue or correction fluid.

The number of marks is given in brackets [] at the end of each question or part question.

Section A to C

Answer all questions.

Write your answers in this booklet.

Section D

Write your answers on the separate Answer Paper.

Begin each section of the question on a new piece of paper.

For Examiner's Use	
Section A - C	[Total: 80]
1	/ 11
2	/ 8
3	/ 12
4	/ 11
5	/ 11
6	/ 6
7	/ 11
8	/ 10
Section D	[Total: 20]
9 / 10	/ 20
Total	/ 100

This document consists of **26** printed pages, including this cover page and one blank page.

Name: _____ Class: 2bi2___ / 2IPbi2___

42

Section A

Answer **all** the questions in this section.

- 1 Fig. 1.1 below shows 2 organelles from an animal cell involved in digestion. Assume that the cell surface membrane is located above the organelles.

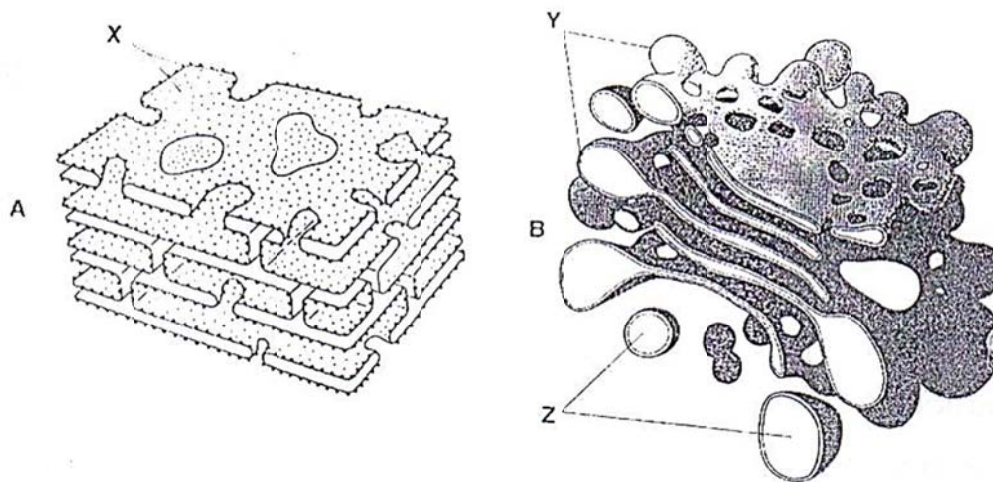


Fig. 1.1

- (a) With reference to Fig. 1.1,

(i) name structures **Y** and **Z** which are associated with organelle **B**.

Structure **Y**: Golgi vesicles / secretory vesicles / cisternae

Structure **Z**: Transport vesicles [1]

(ii) explain the roles and significance of **A** and **B** during protein synthesis .

Ribosomes (Structure X) attach to A, the rough endoplasmic reticulum, and the mRNA is translated to produce a polypeptide that enters the lumen of A

Polypeptide chain might be folded and modified (e.g. secretory proteins are often modified in the lumen by glycosylation) in the lumen of A before being packaged in transport vesicles

B, the Golgi Apparatus, is involved in the modification of the polypeptides synthesised, such as by glycosylation

And sorting the synthesised proteins for transport to specific destinations such as lysosomes, membrane, or secreted out of the cell [4]

(iii) describe how the contents of Y are subsequently secreted out of the cell.

Y, the secretory vesicles, bud off from the trans face of the Golgi apparatus and are transported via microtubules to the cell surface membrane;

The phospholipid membrane bilayers of Y and of cell surface membrane fuses, emptying the contents of Y via exocytosis into the extracellular region; [2]

Fig. 1.2 below shows an electron micrograph of a eukaryotic cell.

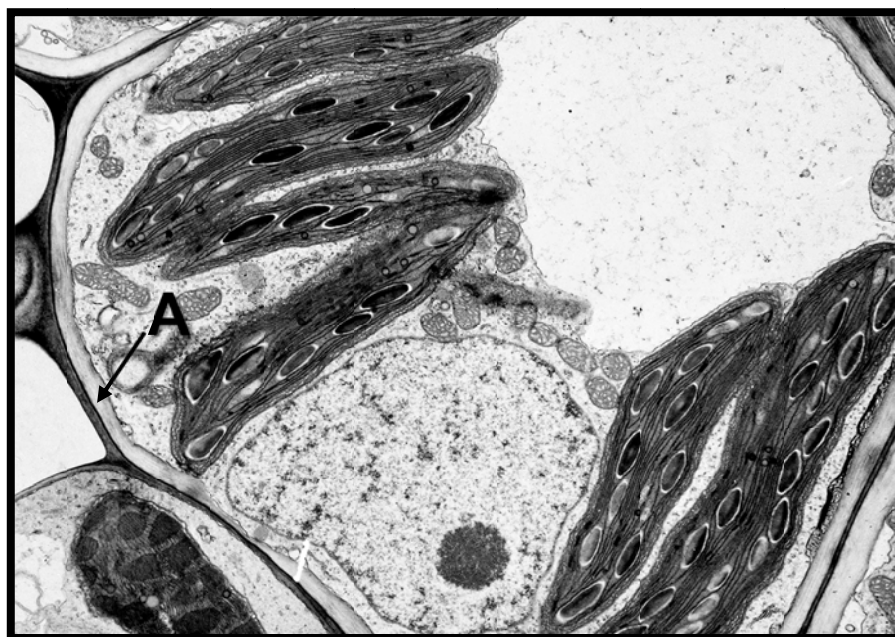


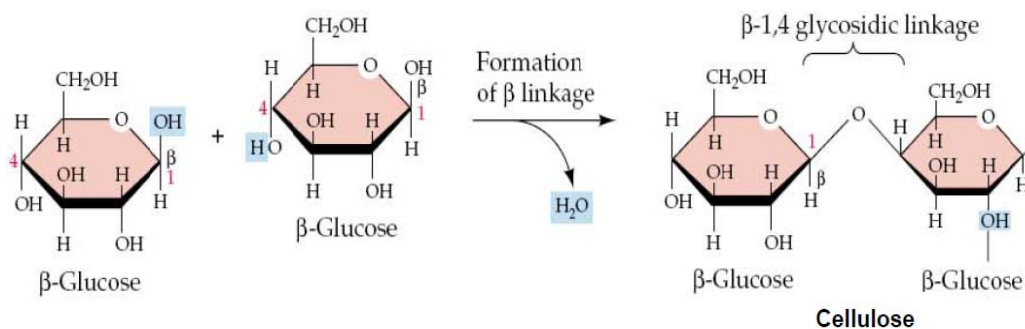
Fig 1.2

- (b) (i) The primary structural component of A is known to be a polysaccharide. Name the monomer of this polysaccharide.

β -glucose

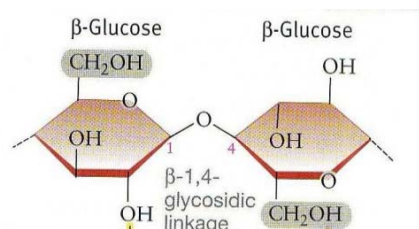
[1]

- (ii) In the space below, draw a labelled diagram to show how two such monomers are joined together.



[1]

Alternatively,



Correct orientation of β -glucose, and labelling of β (1-4) glycosidic bond and H₂O [1]

- (c) Collagen is the most abundant protein found in mammals. Unlike haemoglobin which serves a metabolic function, collagen serves a structural function.

Describe how the molecular structures of collagen and haemoglobin differ.

Features	Collagen	Haemoglobin
1 Shape	Molecule is <u>fibrous</u> in shape.	Molecule is globular in shape.
2 Primary structure	The primary structure consists of about 1000 amino acids residues.]	141 amino acids (α-chain subunit) and 146 amino acids (β-chain subunits)
3 Levels of structure	Tertiary structure is not present / secondary structure is most prominent	Primary, secondary, tertiary and, quaternary structures are present
4 Secondary structure	Secondary structure is a helical polypeptide.	Secondary structure (in both α and β subunits) is primarily α-helical.
5 Quaternary structure	Quaternary structure of tropocollagen consists of 3 polypeptide chains	Quaternary structure of haemoglobin consists of 4 polypeptide chains held together
6 Types of bonds and interactions	Tropocollagen cross-linked by hydrogen bonds to form a triple helix / covalent bonds between tropocollagen	By hydrophobic interactions, ionic and hydrogen bonds.
7 Presence of prosthetic groups	Absence of prosthetic groups	Prosthetic group (haem group) associated with the protein
8 Presence of sequence regularity	Sequence regularity (glycine is the 3 rd amino acid) / regular and repetitive sequence	Absence of sequence regularity / highly irregular and non-repetitive sequence

[2]

9	Specificity of amino acid sequence	May vary slightly between 2 samples of the same protein	• Highly specific and never varies between 2 samples of the same protein
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[Total: 11m]

- 2 The graph below shows the DNA content during nuclear division.

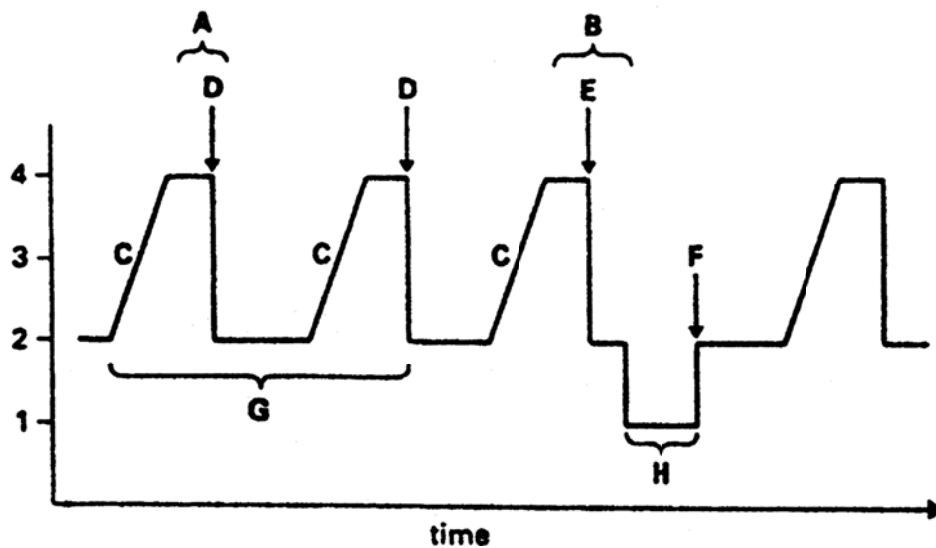


Fig. 2.1

- (a) State the stages **D** and **E** if y-axis of graph indicates DNA content per cell.

Stage **D**: Cytokinesis of mitosis

Stage **E**: Cytokinesis of meiosis I [1]

- (b) Explain the importance of reduction division in meiosis for sexually reproducing organisms.

1 **Meiosis gives rise to haploid gametes / gametes with one set of chromosomes, hence fusion of gametes during fertilisation results in a diploid zygote / enables the diploid condition to be restored upon fusion of gametes**

2 **Hence, maintaining the number of chromosomes / prevents doubling of number of chromosomes in each successive generation**

[2]

Fig. 2.2 shows a linear chromosome undergoing the first round of DNA replication.

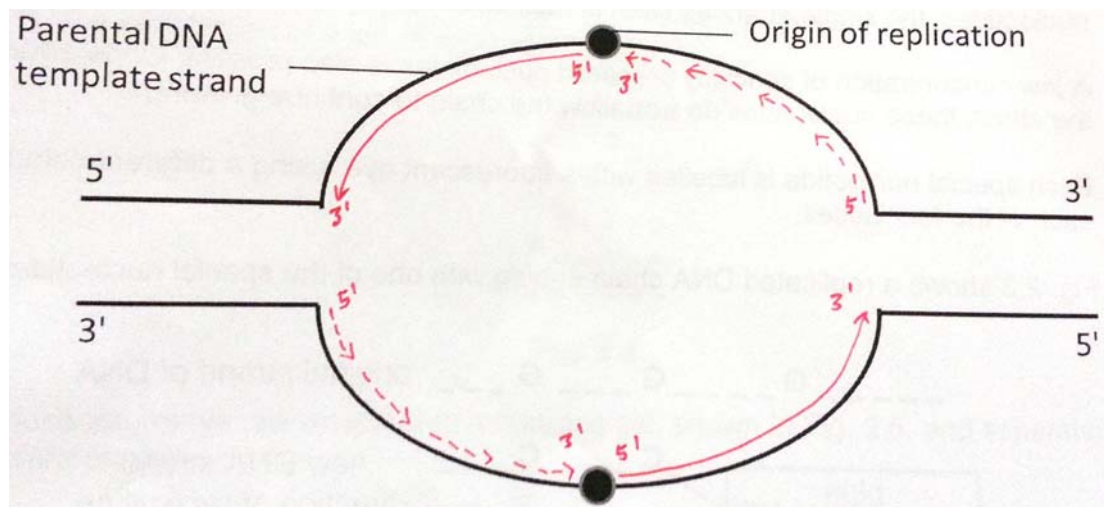


Fig. 2.2

- (c) (i) On Fig. 2.2, draw the direction of DNA synthesis for leading ($\rightarrow$) and lagging strand (- - - - $\rightarrow$).

[1]

(ii) Describe two differences in the formation of the leading and lagging strands.

1. The leading strand is synthesised continuously while the lagging strand is synthesised discontinuously
2. There are presence of Okazaki fragments in the lagging strand while it is not present in the leading strand
3. There is presence of more than one RNA primer in lagging strand while only one primer is needed for synthesis of leading strand
4. The leading strand is synthesised towards the replication fork while the lagging strand is synthesised away from the replication fork

[2]

In order to sequence the DNA of a gene, the DNA is first denatured to separate its two strands. Subsequently, in the presence of a large supply of each of the four nucleotides, the single-stranded DNA is replicated by DNA polymerase.

A low concentration of specially prepared nucleotides is also present. Once added to the chain, these nucleotides do **not** allow the chain to continue growing.

Each special nucleotide is labelled with a fluorescent dye, using a different colour for each of the four bases.

Fig. 2.3 shows a replicated DNA chain ending with one of the special nucleotides.

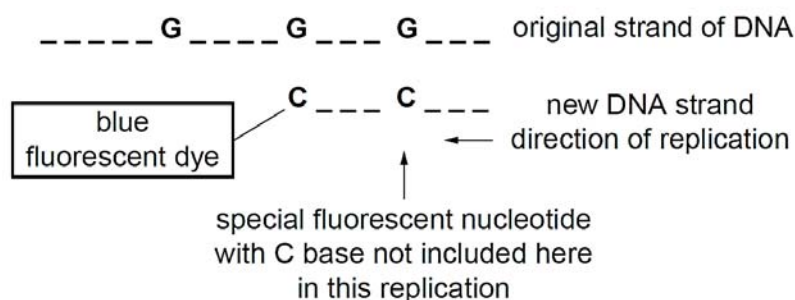


Fig. 2.3

- (d) With reference to Fig. 2.3 and to the information given, suggest why a special nucleotide with a C base was **not** included by DNA polymerase at the first site requiring a C nucleotide.

chance / random ; only present in low concentration (any one) [1]

This method of sequencing a gene produces as many DNA fragments as there are nucleotides in the gene, each fragment differing in length by one nucleotide.

Fig. 2.4 shows part of a set of such fragments.

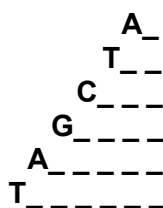


Fig. 2.4

These fragments are loaded onto a agarose gel, shown in Fig. 2.5, and separated by electrophoresis.

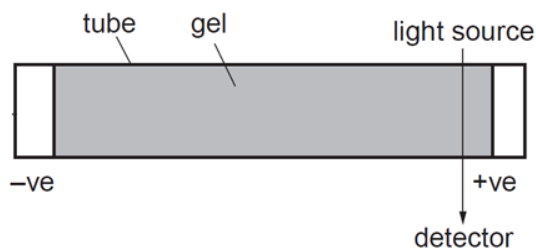
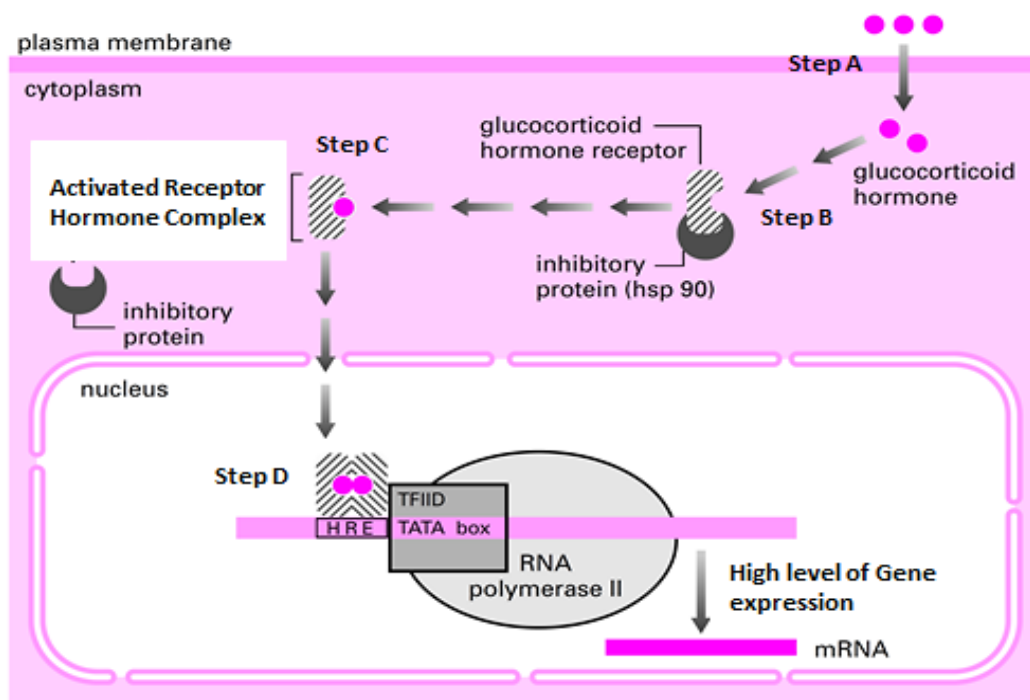


Fig. 2.5

- (e) In what order will the fragments shown in Fig. 2.4 reach the light source and detector shown in Fig. 2.5?

ATCGAT / in order of size starting with shortest [1]
[Total: 8m]

- 3 Glucocorticoids are steroid hormones produced by the adrenal cortex that increase the transcription of several genes important in carbohydrate and protein metabolism. Fig. 3.1 below shows how glucocorticoids can pass through the plasma membrane to enter the cytosol and bind to glucocorticoid hormone receptors. In the absence of glucocorticoid hormone, its receptor remains in the cytosol and is inactive.



Hormone response element (HRE)

Fig. 3.1

- (a) (i) Glucocorticoid hormone receptor is a class of transcription factors. With reference to Fig. 3.1, explain how the receptor hormone complex can be activated.

- Ref to binding of glucocorticoid hormone to inactivated glucocorticoid receptor
- which leads to removal of Hsp90 protein forming the glucocorticoid receptor–hormone complex

[2]

- (ii) Suggest how glucocorticoid hormone receptor is able to bind to the specific region of the DNA known as HRE site, as shown in step D of Fig. 3.1.

- This occurs via Specific DNA - protein interactions which depend upon the sequence of bases in the DNA. [1]
- These DNA - protein interactions are mediated by the following bonds [1]:
 - Hydrogen bonding
 - Ionic interactions: Salt bridges; R groups of protein - DNA backbone interactions
 - Other forces: van der Waals, hydrophobic

[2]

(iii) With reference to Fig. 3.1, explain how high level of gene expression may be regulated by activated receptor hormone complex within the nucleus.

- The glucocorticoid receptor–hormone complex acts as an activator which binds to the HRE which is an enhancer sequence / distal control element.
- The activators bind to certain general transcription factors such as TFIID and mediator proteins / Recruits basal Transcription factors
- Stabilises / increases the binding affinity of RNA polymerase II to TATA Box / Promoter sequence, and increase the rate of formation of stable / stability of transcription initiation complex for RNA synthesis to begin.

[3]

Fig. 3.2 below shows the various steps involved in the processing of primary RNA transcript.

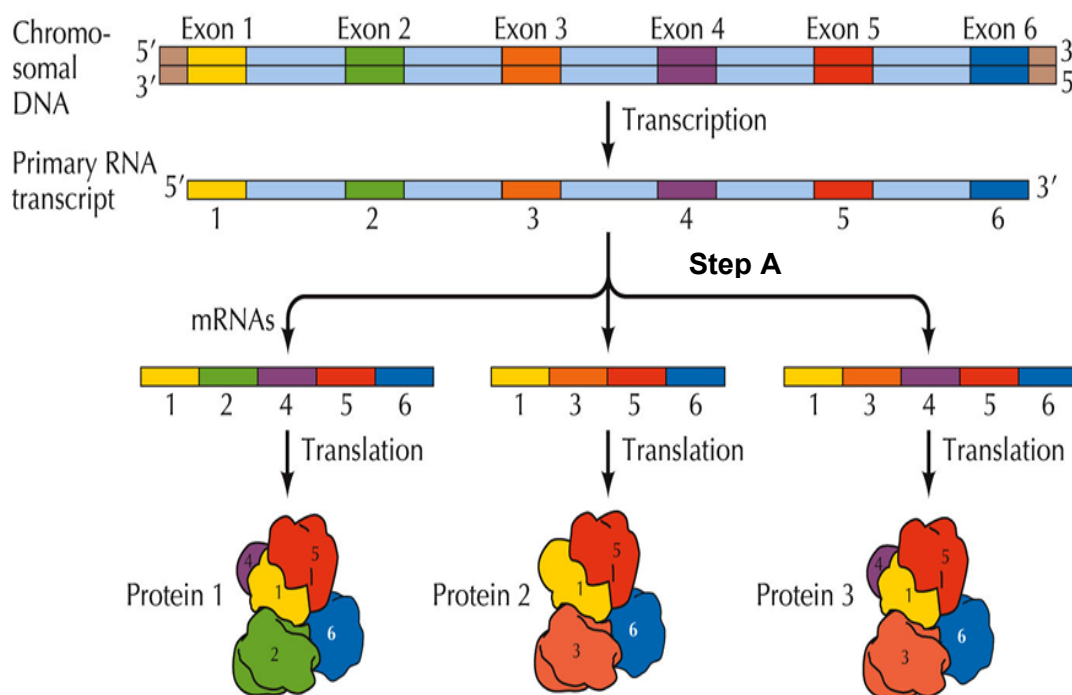


Fig. 3.2

(b) (i) Name step A and explain why it is essential in eukaryotic cells from an evolutionary view point.

1. Step A is Alternative splicing
2. This process allows a single gene to code for different types of proteins / This allows eukaryotic genome to direct the synthesis of many more proteins than would be expected from its fixed protein-coding genes
3. Particular exons of a gene may be included within, or excluded from, the final, processed messenger RNA (mRNA) produced from that gene.
4. Ref to proteins translated from alternatively spliced mRNAs will contain differences in their amino acid sequence and, often, in their biological functions leading to variations to occur in a population. e.g. phenotypes etc.

(Accept either point 2 or 3, but not both together.)

[3]

(ii) Prior to completion of step A, the primary RNA is modified in several important ways. Explain how **one** of these ways helps to regulate gene expression.

Any one of the following:

- Primary mRNA is capped by the addition of 7-methylguanosine cap to 5' end of the transcript to protect the growing RNA transcript from degradation by RNases.
- A poly-A tail is added to the 3' end by the enzyme poly-A polymerase. The poly-A tail has several functions: (1) It protects against RNases/exonuclease and therefore increases the stability of mRNA molecules in the cytoplasm, (2) both it and the 5' cap are required for transit through the nuclear pore from the nucleus to cytoplasm, and (3) it increases the efficiency of translation on the ribosomes.

[1]

(iii) With reference to Fig. 3.3, label the following region (label only once):

- (i) 5' UTR
- (ii) 3' UTR
- (iii) coding sequence
- (iv) promoter

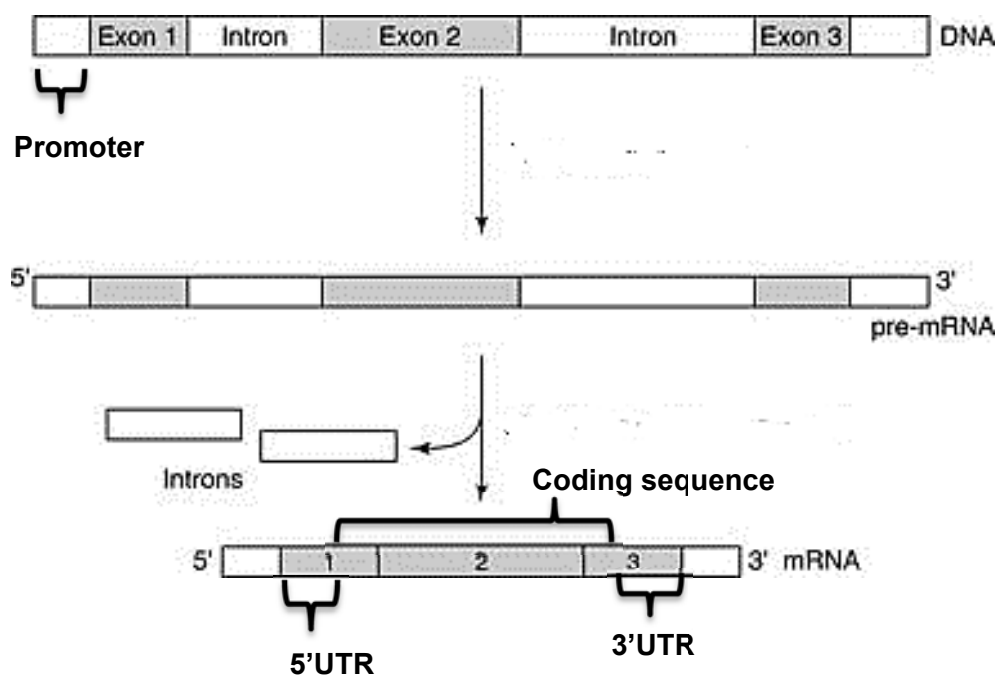


Fig. 3.3

More than 3 correct – 1 mark

[1]

[Total: 12 m]

- 4 Serotonin is a neurotransmitter which is produced by certain neurons in the brain. One of its effects is to increase the activity of sensory neurons in the brain. It also usually improves a person's mood and keeps the person awake. Fig. 4.1 below shows a synapse at which serotonin is the neurotransmitter.

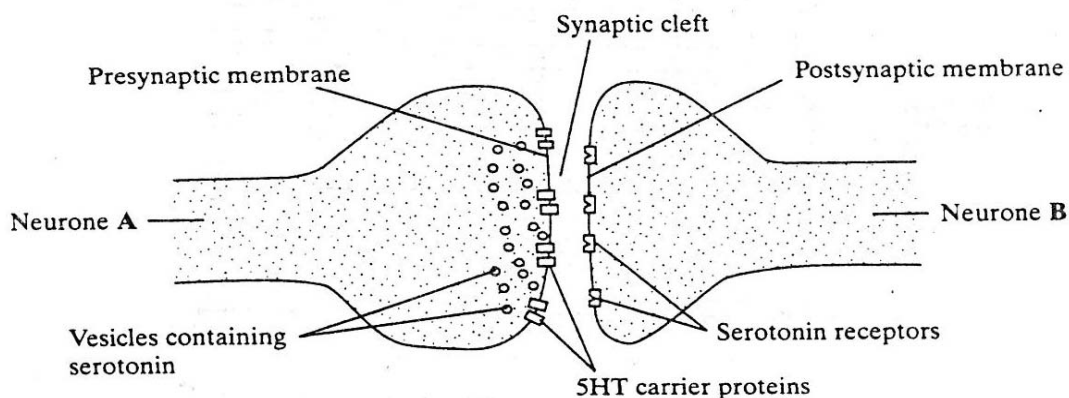


Fig. 4.1

- (a) (i) Explain how the release of the neurotransmitter, serotonin, results in the transmission of an impulse from neuron A to neuron B.

- **Serotonin diffuses across the synaptic cleft to the postsynaptic membrane, and binds to complementary ligand-binding site on serotonin receptors**
R: active site on receptor
- **This causes the opening of ligand-gated Na⁺ channels and influx of Na⁺ ions / increased permeability of the postsynaptic membrane to Na⁺ ions**
R: opening of serotonin receptor
- **Local depolarization of the postsynaptic membrane occurs (i.e. inside becomes more positive)**
- **The build up of local depolarization at the axon hillock due to summation that exceeds threshold potential results in generation of an action potential of neuron B**

[4]

(ii) Serotonin is normally rapidly reabsorbed from the synaptic cleft by 5HT carrier proteins in the presynaptic membrane.

Suggest an advantage of rapidly reabsorbing the serotonin.

- **This prevents overstimulation/continuous stimulation of the postsynaptic membrane by the stimulus from neuron**
- **Allows transmission of impulse to be rapidly terminated**
- **Allows further/faster transmission of another signal across the synapse**
(R: general statements about increasing rate of transmission without reference to a new signal)
- **Allow serotonin to be recycled for rapid formation of new signal**

(any one)

[1]

The pain associated with osteoarthritis can be relieved using transcutaneous electrical nerve stimulation (TENS). It uses electrical impulses to stimulate the nerve endings at, or near, the site of the pain. Self-adhesive electrodes are stuck on the skin and attached to a small, portable power unit.

Fig. 4.2 shows a TENS machine in use.



Fig. 4.2

It is thought that TENS triggers the release of natural painkillers called endorphins, which are similar in shape to painkilling drugs such as morphine.

Fig. 4.3 shows synapses in a pain pathway from a damaged joint to the brain.

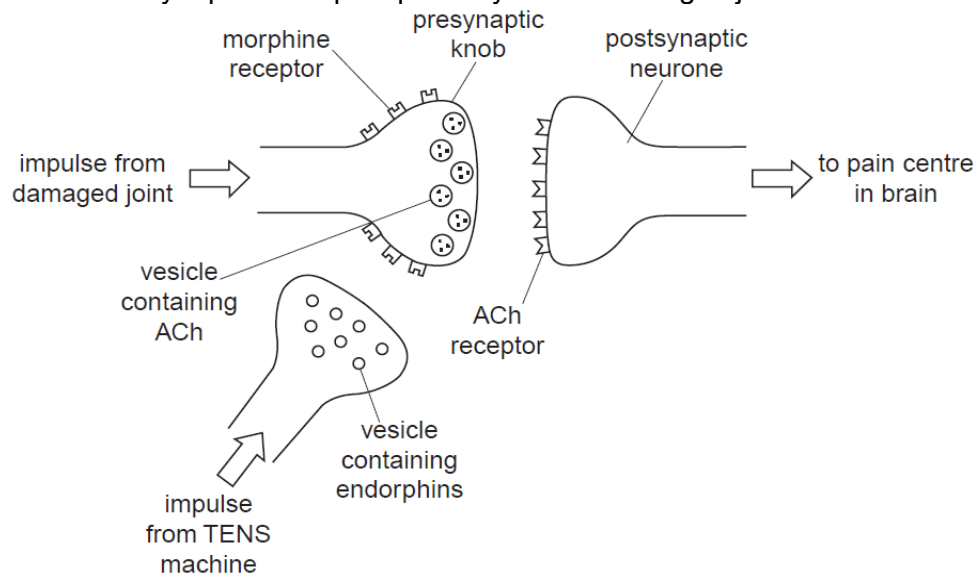


Fig. 4.3

(b) With reference to Fig. 4.3, explain how impulse from the TENS machine may act to reduce pain.

- **impulse from TENS causes release of endorphins, which attach to morphine receptors**
- **slows / stops ACh release ; hence no / less binding of ACh on ACh receptors in postsynaptic membrane**
- **fewer / no action potentials/ impulses, to pain centre (in brain)**..... [3]

- (c) Some synapses in the brain use the neurotransmitter dopamine. After the postsynaptic membrane has been depolarised, dopamine leaves the receptor proteins and moves back into the presynaptic neurone through specific transporter proteins.

Schizophrenia is a condition in which there is a higher than usual concentration of dopamine in certain areas of the brain. The drug phenothiazine has a similar structure to dopamine and is used to treat schizophrenia.

Explain what occurs at the synapse when phenothiazine is used in the treatment of schizophrenia.

- binds to/blocks dopamine receptors on postsynaptic membrane
- prevents depolarization of postsynaptic membrane hence reduces effect of dopamine

R: reduces amount of dopamine

A: bind competitively, binding site, competitive inhibitor. R: active site [2]

- (d) Explain the disadvantage of a signalling cascade.

- Disadvantage: Prone to error; error along any of segment of the cascade would result in undesirable consequences / non specificity of response / wastage of energy.

[1]

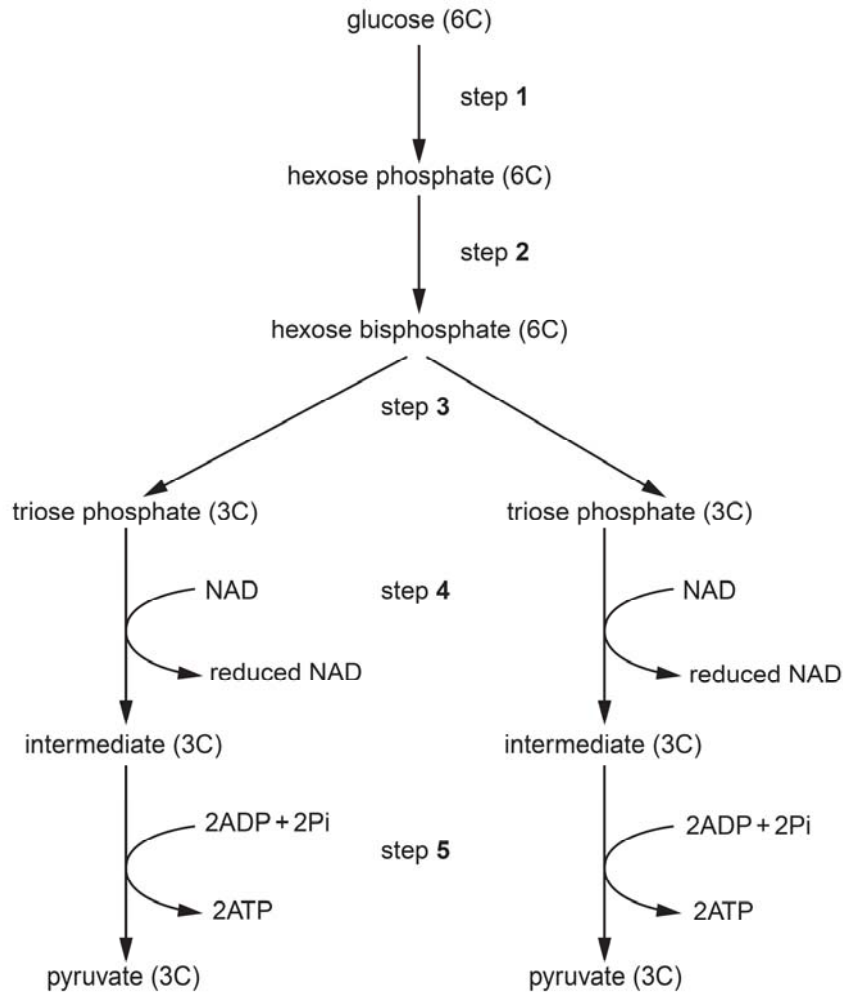
[Total: 11m]

Name: _____ Class: 2bi2__ / 2IPbi2__

17

Section BAnswer **all** the questions in this section.

- 5 During the process of glycolysis, glucose is converted by a series of steps into two molecules of pyruvate. Fig. 5.1 outlines glycolysis.

**Fig. 5.1**

- (a) With reference to Fig. 5.1, state the processes occurring at:

- (i) steps 1 and 2 : **Phosphorylation**
- (ii) step 3 : **Lysis**
- (iii) step 4: **Dehydrogenation or Oxidation**

Any 2 correct: [1M], 3 correct: [2M]

[2]

- (b) Pyruvate can enter a mitochondrion when oxygen is present. Describe what happens to pyruvate in a yeast cell when oxygen is **not** present.
- **Pyruvate is first decarboxylated, releasing 1 molecule of CO_2 in the process.**
 - **The acetaldehyde (ethanal) formed is then reduced by NADH to ethanol. Enzyme: Alcohol dehydrogenase. This reduction regenerates NAD^+ , which is then re-used in glycolysis.**

[2]

- (c) Fig. 5.2 outlines anaerobic respiration in yeast cells. Complete Fig. 5.2 by writing in the missing compounds.

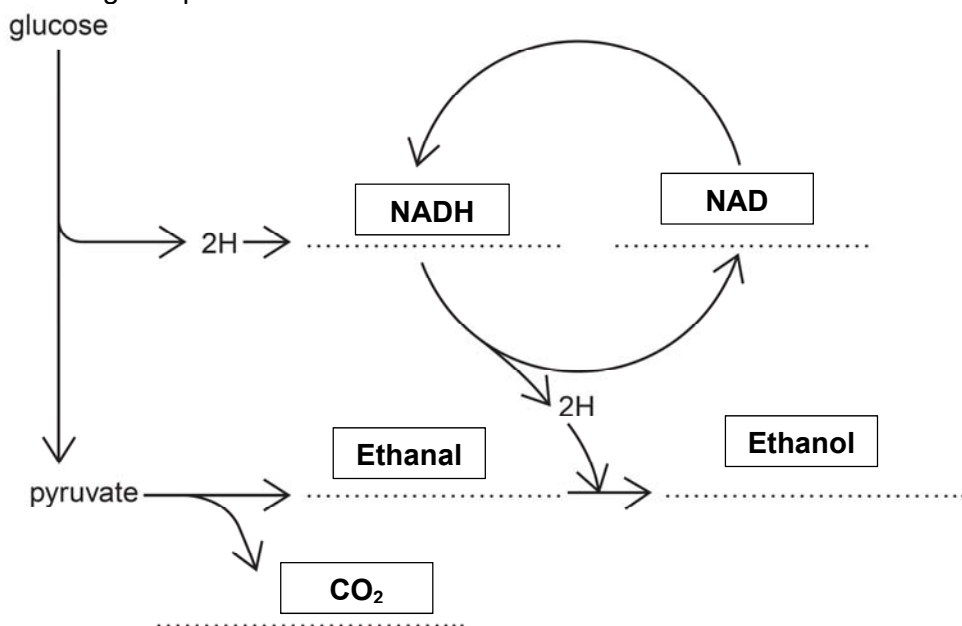


Fig. 5.2

4 Correct: [2M]

3 Correct: [1 M]

[2]

- (d) Describe how anaerobic respiration in mammalian cells differs from anaerobic respiration in yeast cells.

in mammals

- ***lactate produced instead of ethanol [1]***
- ***no decarboxylation / carbon dioxide released and it involves a single step catalysed by lactate dehydrogenase ;***

[2]

Fig. 5.3 shows the changes in the relative concentrations of RuBP and GP produced in the Calvin cycle before and after a light source is switched off. All other conditions are constant.

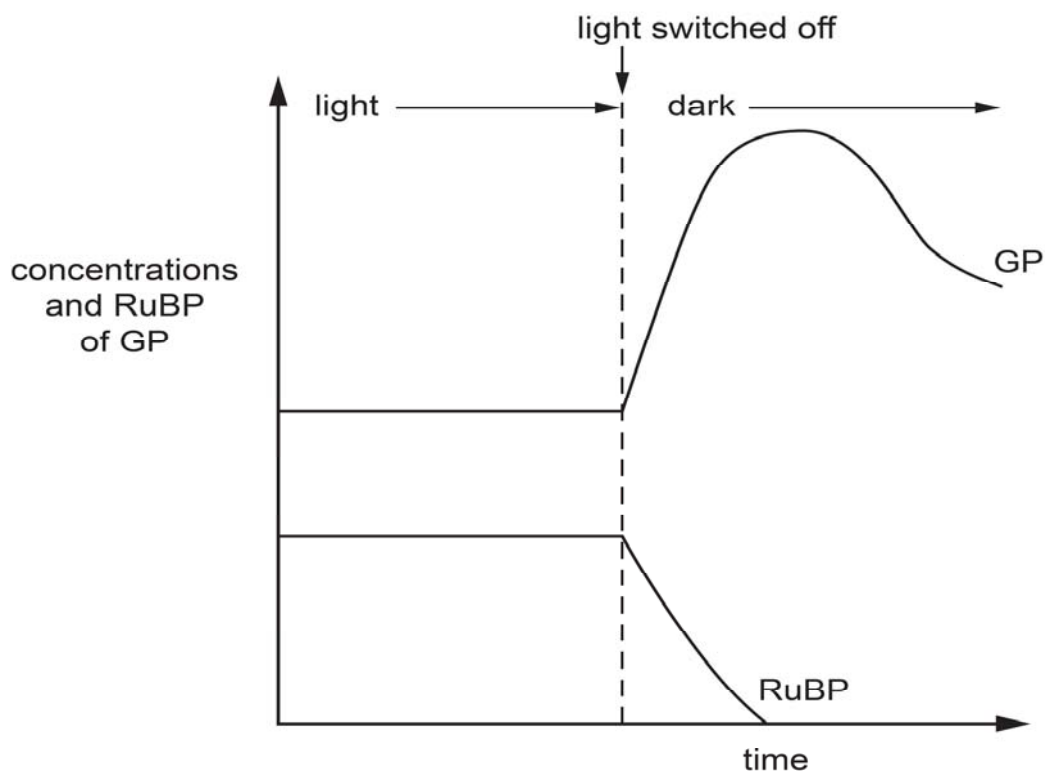


Fig. 5.3

(e) Explain the changes in the relative concentrations of RuBP and GP **after** the light source is switched off.

- **light dependent reaction stops during dark stage, no ATP and NADPH being produced. RuBP is not regenerated ; [1]**
- **light independent reaction / Calvin cycle will still continue to occur the RuBP present will still be converted to GP using existing ATP and NADPH present in the system (produced during light stage). This will continue until ATP and NADPH are fully used up ; [1]**
- **GP will be converted to Triose Phosphate which will then be used to make hexose as shown by the reduction in the graph [1]**

[3]

[Total: 11m]

- 6 It was observed that when a pure-breeding strain of squash that produces disc-shaped fruits was crossed with a pure-breeding strain having long fruits, all F₁ had disc-shaped fruits.

Once fully grown, the resulting F₁ plants were self-pollinated. The phenotypes of the fruits produced (shown in Fig. 6.1) were recorded in the following proportions:

Long	32
Sphere	178
Disc	270

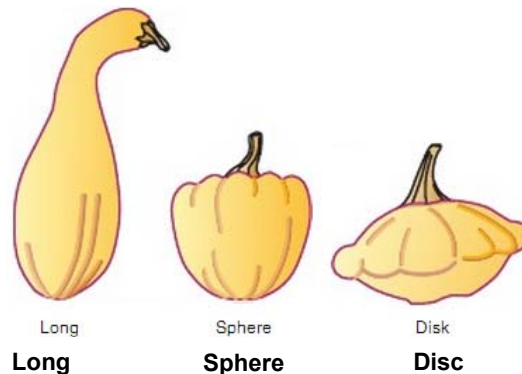


Fig. 6.1

- (a) Draw a genetic diagram in the space below to show the cross described and show the genotypes of the P, F₁, and F₂ generations. [4]

The alleles are represented by the following symbols:

A represent the dominant allele for enzyme production

a represent the recessive allele for enzyme deficiency

B represent the dominant allele for enzyme production

b represent the recessive allele for enzyme deficiency

Squash with genotype A₋B₋ is disc-shaped, genotype

Squash "aabb" - long-shaped, and with genotypes "A₋bb" and "aaB₋" - ball-shaped.

Hint: Note that the F₂ are in an approximate 9:6:1 ratio. This suggests a di-hybrid cross in which A₋bb has the same appearance as aaB₋. [Recall: 9:3:3:1 ratio]

Let the disc phenotype be the result of A₋B₋ and the long phenotype be the result of aabb.

Summarise in the form of the table:

Genotypes	Phenotypes	Explanation	F ₂ Ratio
9 A ₋ B ₋	Disc Shaped	Dominant A and B will express functional enzyme A and B to produce disc shape.	9
3 A ₋ bb	Sphere Shaped	As long as any one of the enzyme is functioning, Sphere shaped is produced.	6
3 aaB ₋	Sphere Shaped		
1 aabb	Long Shaped	No functional enzymes, long shape fruit is produced	1

Name: _____ Class: 2bi2___ / 2IPbi2___

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Section C

Answer **all** the questions in this section.

- 7 (a) In bacteria, it is common for two or more genes to be arranged together in an operon.

(i) Briefly discuss the arrangement of genetic sequences within an operon.

Any 1 from below (max. 1 mark):

- **Flanked by a promoter to signal the beginning of transcription and a terminator to signal end of transcription**
- **An operator site, is usually present, to serve as binding site for regulatory proteins called activators or repressors**

1 from below:

- **Two or more structural genes that encode different proteins are found between these two sequences**

[2]

(ii) State the biological advantage of an operon organization.

- **Co-ordinately regulate a group of genes whose encoded proteins have a common function to reduce wastage of resources / response to environmental changes quickly**

[1]

- (b) The sequential use of two sugars by a bacterium is known as diauxic growth. It is a common phenomenon among many bacterial species. When glucose is one of the two sugars available, it is typical that the bacterium metabolises glucose first, and then a second sugar after the glucose has been used up. Among *E. coli* and related species, diauxic growth is regulated by intracellular cAMP levels and the catabolite activator protein (CAP).

(i) Summarise the effects of glucose and lactose on the ability of the lac repressor and the cAMP-CAP complex to regulate the *lac* operon.

- **Lactose present, operon is switched on, as the repressor protein is removed from the operator site**
- **Glucose present, operon is turned off due to the dominating effect of the lac repressor protein**
- **Lactose and glucose present, the expression of the *lac* operon is greatly decreased, the lac repressor is removed from operator site, and the catabolite activator protein is not bound to the CAP site, reducing RNA polymerase binding to begin transcription**

[3]

You have isolated a mutant strain of *E. coli* in which the *lac* operon is constitutively expressed. To understand the nature of this defect, you create a merozygote (diploid cell) in which the mutant strain contains an F' factor with a normal *lac* operon and a normal *lacI* gene. You then compare the mutant strain and the merozygote with regard to their β -galactosidase activities in the presence and absence of lactose.

You obtain the following results:

	Addition of lactose	Amount of B-galactosidase (percentage of mutant strain in the presence of lactose)
Mutant	No	100
Mutant	Yes	100
Merozygote	No	100
Merozygote	Yes	200

Fig. 7.1

(ii) Explain the nature and effect of the defect in the mutant strain.

- **Mutation in the operator site**
- **Repressor protein cannot bind to prevent lac operon expression in the absence of lactose**

[2]

(c) Human Immunodeficiency Virus (HIV) is the cause of Acquired Immunodeficiency Syndrome (AIDS).

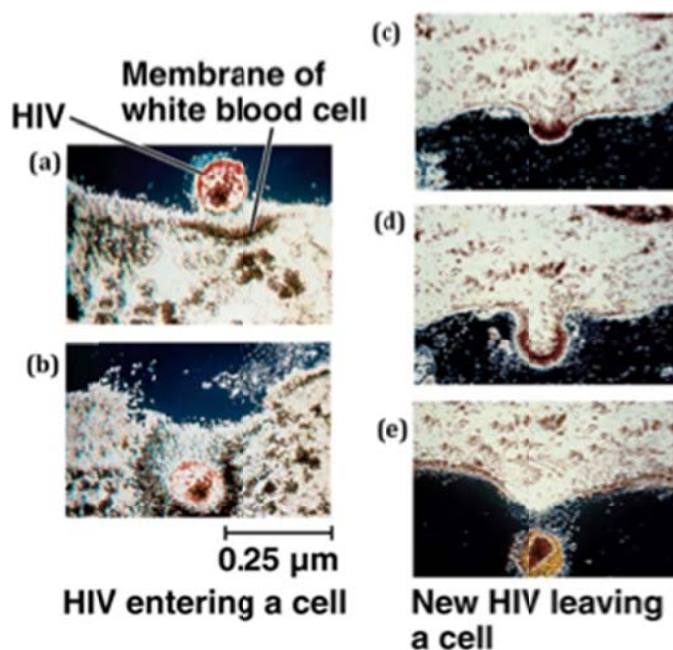


Fig. 7.2

(i) Describe the events occurring during (a) and (b).

- **Glycoprotein (gp120) on the viral envelope bind to specific cell surface receptor molecules (CD4 surface receptors) on the host cell (CD4+ T cells)**
- **Fusion of HIV membrane envelope with CD4+ T cell surface membrane**

[2]

(ii) Although there are medications to prevent the replication of HIV, explain why there is no full cure.

- **Integration of HIV genome into CD4+ T cells**
- **Replication of virus is prevented through medication, if and when patient stops taking medication, viral genome can activate replication**

[1]

[Total: 11m]

8 Mammoths are extinct mammals related to elephants. About three million years ago, the ancestors of mammoths migrated from Africa into Europe and Asia. There, about 1.7 million years ago, the steppe mammoth (*Mammuthus trogontherii*) evolved and became adapted to the cooler conditions. Then, about 700 000 years ago, as the climate changed and the Arctic became much colder, the woolly mammoth (*Mammuthus primigenius*) evolved. Woolly mammoths showed a number of obvious adaptations to reduce heat loss, including thick fur, small ears and small tails.

(a) Explain how the evolution of the woolly mammoth of small tail, small ears, and thick fur came about.

- *individuals varied (in their phenotypes);*
- *(phenotypic variation) caused by, genetic variation / mutation;*
- *change in, selection pressure / environmental conditions;*
- *idea that variation increases the chance of some individuals surviving / AW;*
- *named adaptation explained; e.g. better insulation / smaller surface area to volume*
- *survivors breed;*
- *passed on alleles to offspring;*
- *changed allele frequency (in population);*

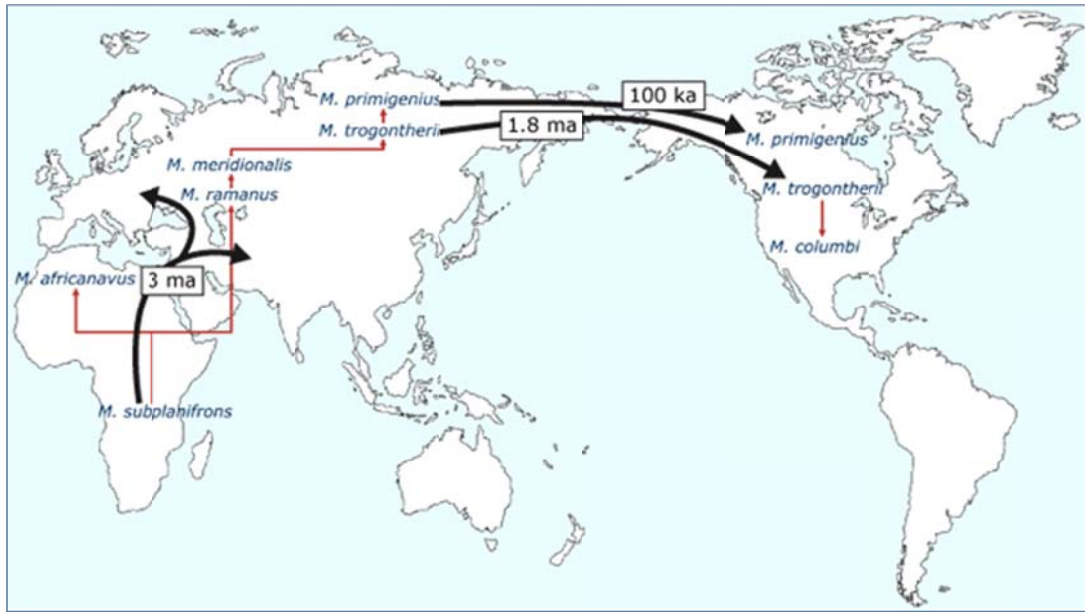
[4]

(b) A frozen 43 000 year old woolly mammoth was found in Siberia. Its DNA was extracted and sequenced. The sequences of the genes coding for the α and β chains of haemoglobin were compared with those of modern Asian elephants. State two advantages of molecular methods in the study of evolutionary lineage.

- *Molecular data are readily available for analysis and can be easily interpreted;*
- *Less difficulty in defining characters for molecular data as compared to anatomical features;*
- *A large majority of extinct organisms do not have anatomical records for comparison*
- *allow comparison of neutral changes in organisms which might have been passed down through evolution, not only adaptive changes*

[2]

The map in Fig. 8.1 shows the mammoth migration and evolution around the world. Migratory events, either in the form of range expansion (natural movement away from parents) or jump dispersal (traveling long distance across inhospitable habitats), may lead to genetic isolation, thus facilitating speciation.



* Legend: ma – million years ago; ka – thousand years ago

Fig. 8.1

- (c) (i) Suggest, with reason, which mode of speciation the mammoths underwent.

Allopatric speciation. The ancestral population migrates across continent, thus presence of geographical barrier.

[1]

- (ii) State and describe two possible causes of sympatric speciation.

- ***Temporal isolation; species reproduce in different seasons or at different times of the day***
- ***Behavioural isolation; species differ in their mating rituals***
- ***Mechanical isolation; structural differences between species prevent mating***
- ***Gametic isolation; gametes of one species function poorly with gametes of another species or within the reproductive tract of another species***

[2]

- (iii) Suggest why biological species concept cannot be used to classify mammoths into the different species.

- ***Non-living species, cannot reproduce and provide fertile offsprings***

[1]

[Total: 10m]

Section D

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

BEGIN EACH SECTION OF THE QUESTION ON A NEW PIECE OF ANSWER PAPER.

9 (a) Explain how fluidity of biological membranes can be maintained.

[6]

- Definition of fluidity: phospholipids and proteins free to move within layer / transversely
- Membrane fluidity can be regulated by regulating the level of cholesterol and phospholipids with unsaturated hydrocarbon tails in the membrane
- Membranes freeze at a lower temperature if it has a higher proportion of phospholipids with unsaturated hydrocarbon tails and cholesterol (Accept reverse argument)
- Unsaturated hydrocarbon tails have kinks that keep the phospholipid molecules in the membrane from packing close together at lower temperatures, enhancing membrane fluidity (Accept reverse argument)
- Cholesterol prevents close packing of phospholipids in the membrane, preventing membrane from freezing at lower temperatures
- Membrane prevented from being overly fluid / integrity of the membrane is maintained at higher temperature as cholesterol restricts phospholipid movement through its interactions with phospholipids
- stabilizing the lipid bilayer through hydrophobic interactions / Van der waal forces
- Length of fatty acid tails also help with the maintenance of membrane fluidity, since longer tails will result in greater amount of hydrophobic interactions between the tails and hence, decrease in membrane fluidity (accept reverse argument);

(b) Describe the main ways in which a globular protein differs from DNA.

[8]

Feature	DNA	Globular Protein
Monomers	The basic unit is deoxyribonucleotide, which comprises a deoxyribose, a phosphate group and a nitrogenous base.	The basic unit is amino acid. Amino Acid Structure The diagram shows the general structure of an amino acid. A central carbon atom (the α-carbon) is bonded to four groups: a hydrogen atom (H), an amino group (H-N), a carboxylic acid group (C=O and OH), and a side chain (R). Labels with arrows point to the 'Amino Group', 'Side Chain', and 'Carboxylic Acid Group'.
Types of monomers	There are four different bases, A, T, C and G.	There are twenty different amino acids.
Elements present	Consists of C, O, N, H, and P atoms.	Consists of C, O, N and H atoms. S atoms may also be present.

Linkage between monomers	Phosphodiester bonds join deoxyribonucleotides together.	Amino acids are joined by peptide bonds.
Structural features	Two polynucleotide chains are wound together to form a double helix.	One or more polypeptide chains are folded to form a spherical shape.
Bonding between chains	Hydrogen bonds exist between nitrogenous bases of two DNA strands of a double helix. A and T are held by two hydrogen bonds while C and G are held together by three hydrogen bonds. Hydrophobic interactions occur between stacked bases.	Polypeptide chains are involved in inter-chain interactions between R groups of the amino acids in the different subunits such as hydrogen bonds, ionic bonds, disulfide bonds and hydrophobic interactions.
Location in the cell	DNA is present in the nucleus of eukaryotic cells, as well as in mitochondria and chloroplasts.	Globular proteins may be present throughout the cell.
Synthesis and template used	DNA is synthesised by semi-conservative replication in the nucleus, using the original DNA as a template.	Synthesis of globular proteins occurs in the cytoplasm, on ribosomes, using mRNA as a template.
Role(s)	DNA is the hereditary material of all living organisms.	Globular proteins serve metabolic roles in cells, as enzymes, hormones etc.
Overall charge	Negatively charged due to the phosphate groups in the backbone.	Variable overall charge depending on the nature of amino acids making up the protein.

- (c) Explain, with examples, how anatomical and embryological homology supports Darwin's Theory of Natural Selection. [6]

Anatomical homologies [Max 2 pts]

- Anatomical homologies are structures such as bones and other physical structures in two species that are derived from a common ancestor even though the structures may have different functions
- An example of anatomical homologies is the similarity in the number and arrangement of bones in the forelimbs of mammals. [Other examples of anatomical homologies are the vertebra column, presence of hair, mammary gland]

Embryological homologies [Max 2 pts]

- Embryological homologies are the similar developmental patterns and developmental features of embryos belonging to organisms that are derived from a common ancestor.
- An example of embryological homologies is the presence of pharyngeal pouches / post-anal tail in both chick embryo and human embryo.

Supporting Darwin's Theory [Max 2 pts]

- Homologies provide the basis of comparison between species to determine their evolutionary relationship. Generally, the greater the amount of homology, the more closely related the species.
- Non-homologous (analogous) structures do not provide such a basis to determine evolutionary relationship because analogous structures are derived from different ancestors / evolved separately in different lineages, although they may have a similar function
- Homologies suggest common ancestry, for example, mammalian forelimb is a homologous feature, with different forms in different species. This suggests that they descended from a common ancestor, which had a basic form of the pentadactyl limb.
- Homologies show "descent with modification": Comparisons of homologies (anatomical, embryological and molecular) between species show how an ancestral structure or gene in a population may have been modified in descendent species through the process of natural selection.
- Modified traits that best fit the species to the particular environment / are selectively advantageous such traits will increase in frequency over many generations.

[Total: 20]

- 10 (a) Describe the functions of common proto-oncogenes and tumour suppressor genes and explain how loss of function mutation and gain of function mutation can contribute to cancer. [6]

Definition of Proto-oncogenes, e.g. ras gene: [Max 1m]

- B1. Proto-oncogenes codes for protein products that **stimulate NORMAL cell division** and cell growth. These proteins are involved in signal transduction pathway.
- B2. example **ras gene which codes for a G protein** ;relays a signal from growth factor receptor (GPCR) which results in **activation/stimulation of cell cycle**;

Mechanism of the Mutation: [Max 1m]

- B3. Gain of function mutation is **dominant** as **only one allele** of the *ras* gene (proto-oncogene) needs to be converted to an oncogene.
Note: Reject: mutation in only one ras gene → you are imply that there is more than one gene/ The use of the word copies instead of alleles
- B4. This is sufficient for the **over-expression of proteins** which can lead to excessive cell division → *cancer Phenotype*;

Outcomes: Any 1

- B5. Gain of function mutations in the *ras* gene can lead to:
- production of a **hyperactive Ras protein** (even in the absence of a signal);
 - increase expression** leading to **excess amounts of Ras protein** (G protein);
- B6. **kinase cascade is triggered continuously**; even in the absence of growth factor, resulting in **excessive cell division**;

Definition of Tumour Suppressor genes-, e.g. p53 gene: [Max 1m]

- B7. Tumour suppressor genes e.g. p53 codes for p53 protein which is a **transcription factor**; that promotes the expression of genes / synthesis of p53 proteins. Functions of p53 proteins:
- p53 will **inhibit cell division** when necessary during DNA damage;
 - activate other genes (e.g. p21)** that synthesise proteins that repair damaged DNA
 - Gene products may **trigger cell death** (apoptosis) if DNA damage is **beyond control**.

Mechanisms of the mutation: [Max 1m]

- B8. Loss of function mutations that **decrease the expression** of the gene / cause **protein to be less or inactive**, thus giving rise to a mutated tumour suppressor gene.
- B9. Most loss-of-function mutations are recessive, because in a heterozygote the second chromosome copy (allele) carries an **unmutated version** of the gene, **sufficient to code for fully functional proteins** whose presence compensates for the effect of the mutation.
- B10. mutations in **both p53 alleles of the tumour suppressor gene** are required to cause the cancer phenotype;

Outcomes: Any 1 [Max 1m]

- B11. mutation of *p53* results in failure to express genes such as
- p21* or genes involved inhibition of cell cycle, hence **cell cycle cannot be inhibited / suppressed during DNA damage**;
 - genes, the protein products of which are **involved in DNA repair**, hence cancer-causing mutations cannot be repaired;
 - suicide genes which are activated for cells whose DNA damage is beyond repair, hence **apoptosis is inhibited**;
- B12. **absence of inhibition / suppression** of cell cycle result in **uncontrolled cell growth**;

Summary point

- B13. **The accumulation of mutations including the appearance of 1 active oncogene and several tumour suppressor genes is required for cancer to develop.**

- (b) Explain how a change in the **sequence of the DNA nucleotide** may affect the **amino acid sequence in a protein**, and hence the **phenotype** of the organism. [8]

Definition [1]

- B1. Change in **coding sequence of DNA nucleotides** can take place due to **addition, deletion or substitution** of DNA nucleotides.
- B2. results in a frameshift in the nucleotide sequence of the allele or different codon sequence in a DNA molecule.

Addition or deletion of sequence [1]

- B1. **Deletion and insertion** mutations may have distinct effects on gene coding sequence. This could lead to **loss or gain of amino acids** having varying effects on the function of the encoded protein.
- B2. If the number of deleted or inserted nucleotides is **three or a multiple of three**, deletions or insertions of this type will be less impactful. It will have an impact if, amino acids involved in an enzyme's active site are lost, or if an insertion disrupts an important secondary structure in the protein.
- B3. If the number of deleted or inserted nucleotides is **not three or a multiple of three** then a frameshift mutation results, all of the codons downstream of the mutation being taken from a different reading frame from that used in the unmutated gene. This will have a **significant effect on the protein function**, because a greater or lesser part of the mutated polypeptide has a completely different sequence to the normal polypeptide.

Base Pair substitution: [1]

- B1. Single base / nucleotide substitution within the gene;
- B2. This will lead to a change in the codon, hence may lead to a change in the amino acid.

B3. Outcomes of base pair substitution is more significant if it happens at the first and second base pair of each codon (3rd base pair being “wobble” hence might not lead to a change in the amino acid the codon is coding for.)

Outcome of change in nucleotide sequence: [2]

B1. Changes in DNA sequence due to mutations will cause different amino acids sequence being incorporated into the polypeptide chain.

B2. As the R groups of the original amino acids are involved in forming bonds that hold the protein in its precise shape,

B3. The altered amino acid sequence will change the shape/ molecular configuration of the protein molecule.

B4. Since its function depends on its shape, the protein becomes non-functional/ altered in function and hence the phenotype of the organism is altered.

Example of how of an organism may be affected: Sickle cell anaemia [3] or Insulin protein e.g. diabetes mellitus

B1. One example would be in Sickle patients, where thymine substitutes with adenine within the gene coding for β -globin chain;

B2. a codon in mRNA is **changed** from GAA to GUA; Results in a **change of amino acid** with a **different R group**

B3. Glutamic acid is replaced by **valine**; Glutamic acid carries a negative charge & is **polar/ hydrophilic**, while **valine** is **nonpolar & hydrophobic**;

B4. Primary structure of β -globin chain is **altered**, leading to **change in folding and coiling** of the polypeptide; this lead to formation of HBS

B5. HbS molecule **aggregates/ crystalises under low oxygen conditions**, so is **inefficient in carrying oxygen**. This will cause **red blood cells to change from circular biconcave shape to sickle shape**;

B6. Sickle shaped red blood cells can **obstruct blood vessels and haemolyse readily**, leading to **phenotype of anaemia**;

(c) Describe how the structures of RNA are adapted for translation.

[6]

Messenger RNA (mRNA)

- Single-stranded; allows tRNA with complementary anticodon sequences to bind
- One codon codes for one amino acid; the order of the codons on the mRNA specify the order in which amino acids are sequenced to form a polypeptide.
- mRNA (5'UTR) has ribosome attachment site;
- Codons are non-overlapping
- Has spliceosome recognition sites to allow for excising of introns;
- Mature mRNA consists of exons/ no introns;
- Has START/ INITIATOR (AUG) codon and STOP (UAA, UGA, UAG) codons;
- 5' 7-methyl guanosine cap and 3' poly-A tail to ensure mRNA stability;

- Complementary base pairing between codon and anticodons; (award only once in either mRNA or tRNA discussion)
- (2 marks max)

Transfer RNA (tRNA)

- Has CCA site at 3' end for attachment to amino acid/ has amino acid attachment site;
- At least 20 different tRNAs - one for each amino acid
- Has anticodons/ every codon has a corresponding anticodon/ specific triplet base sequence complementary to the codon at a region of the folded tRNA molecule which in turn carries the corresponding amino acid.
- Has shape complementary to amino acyl tRNA synthetase for activation of amino acid;
- Internal hydrogen bonds to form clover leaf shape/ have specific 3D configuration/ stabilize the molecule;

(2 marks max)

Ribosomal RNA (rRNA)

- A structural component of ribosome;
- Due to hydrogen bonding/ complementary base pairing within;
- Gives rise to a variety of ribosomal types (e.g. 70S, 80S, 50S, 30S, etc);
- The rRNA (large subunit of ribosome) is important for the peptidyl transferase activity that links amino acids together

(2 marks max)

[Total: 20]

---The End---